GRADUATION PROJECT JULY 2026

Non-Invasive Blood Glucometer

PROJECT TEAM

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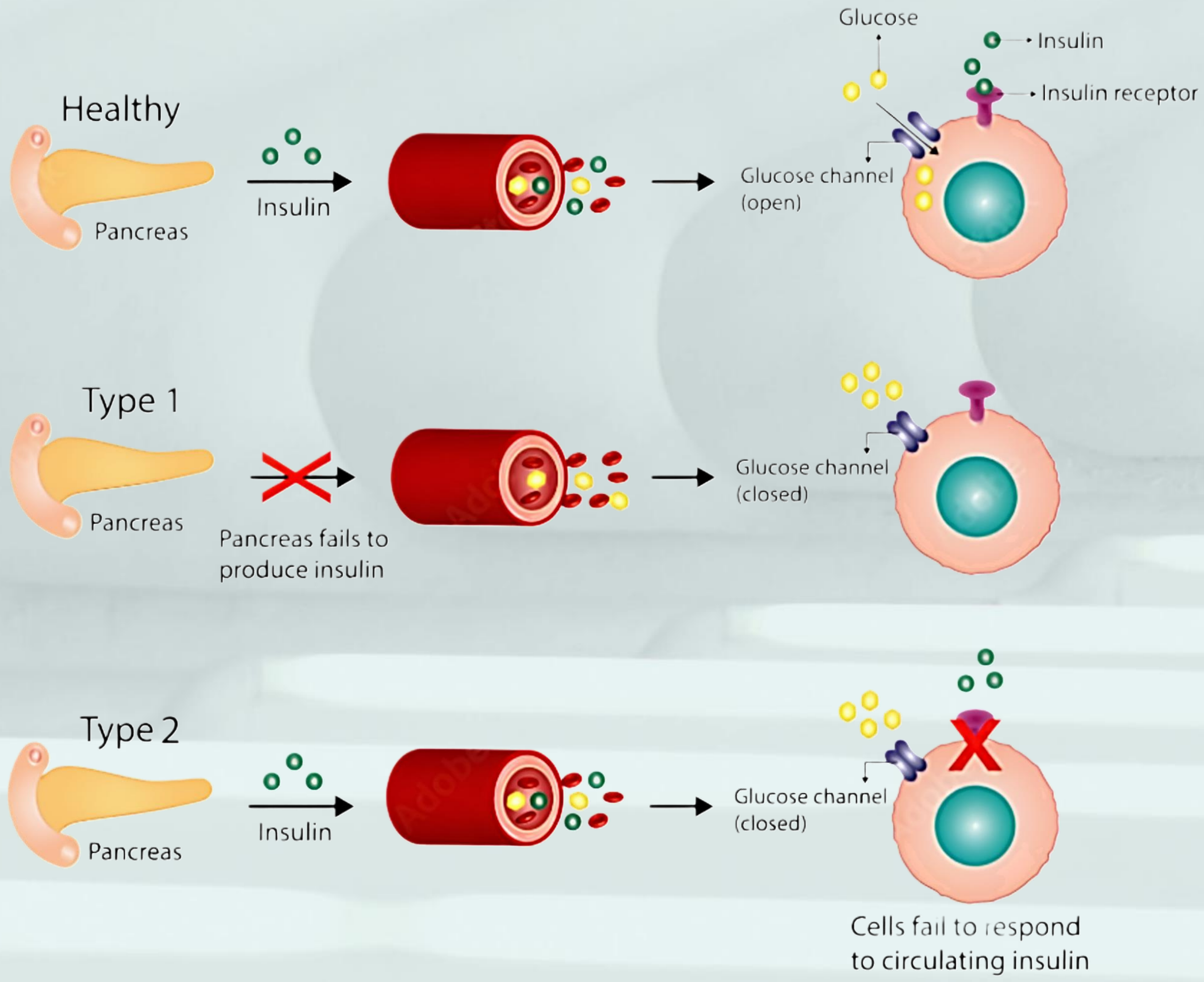
CONTENT

01

Introduction

Diabetes

- This condition arises when the body either **does not produce enough insulin** or **cannot** effectively **use** the insulin it produces.
- There are several types of diabetes, the most common being:
  - Type 1 diabetes
  - Type 2 diabetes
- **1 in 9** people are living with diabetes worldwide, and **its rising**.



How Does People With Diabetes Keep Track Of Their Blood Glucose Levels?

BEWARE

Random high glucose reading **does not confirm diabetes**, diagnosis requires proper medical testing.

Glucose Meters

Invasive Glucose Meters

1

Pricking the finger several times A day

Tissue damage

Reliable

Painful

Risk of infection

Glucose Meters

Continuous Glucose Monitors (CGMs)

2

Batch inserted under the skin

Have a slight time lag

Expensive

Patch replaced every 10–15 days

Risk of infection

Objective

Non-Invasive Blood Glucometer

3

No finger pricking, test strips, or patches.

Comfortable

Cost-efficient

Continuous monitoring

However, it is still **under development**, so its generally less accurate than invasive or minimally invasive devices,

but it’s showing **significant promise**.

02

How Will It Work?

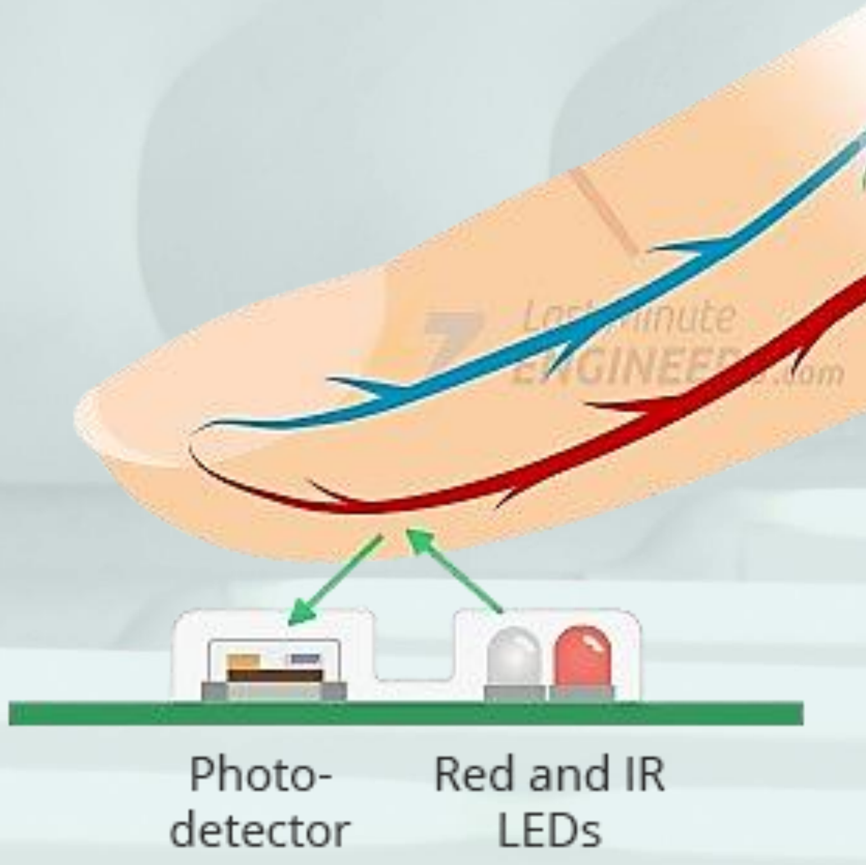
Principle

- Photoplethysmography **(PPG) signals**.
- Simple yet powerful optical technique used to detect **blood volume changes** in the microvascular bed of tissue (e.g., finger).

I. An **LED** emits light (**Red** at 660nm, **IR** at 880nm) into the tissue barrier.

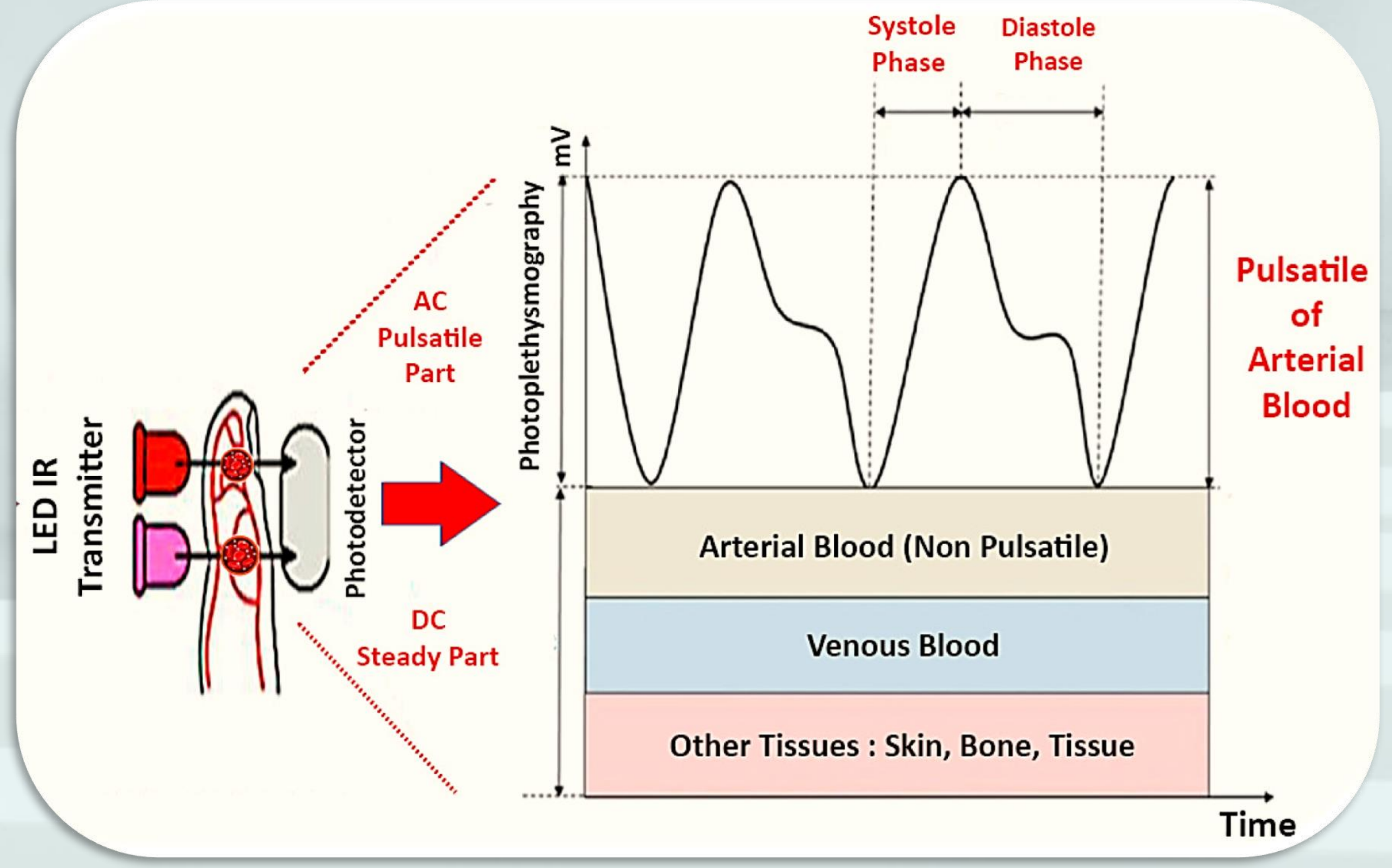
II. Biological components (skin, bone, muscle, and blood) **absorbs** the light.

III. A highly sensitive **photodetector** captures the remaining scattered or reflected light.



PPG Signal

- PPG signal consists of two main components:
  - **AC** Component (Pulsatile): Driven strictly by the arterial blood inflow of the **cardiac cycle**. Contains markers for **heart rate**.
  - **DC** Component (Baseline): Represents constant light absorption by **bones**, **structural tissue**, and non-pulsatile blood.



Bio-Optical Bridge to BGL

- Chemical changes in blood glucose levels directly alter the physical
  - **Optical (DC Alteration)**
    - Changing how light **scatters** through tissue and modifying the static baseline signal.
  - **Mechanical (AC Alteration)**
    - Increasing blood **viscosity** and arterial stiffness and flattens the systolic wave slopes.
- behavior of the peripheral tissue bed (blood).
- That’s why its already used in **researches** in:
  - **Classifying diabetes.**
  - **Predicting BGLs**.

Workflow

NIR → PPG Signal → Preprocessing Feature Extraction → ML Model → Glucose Level

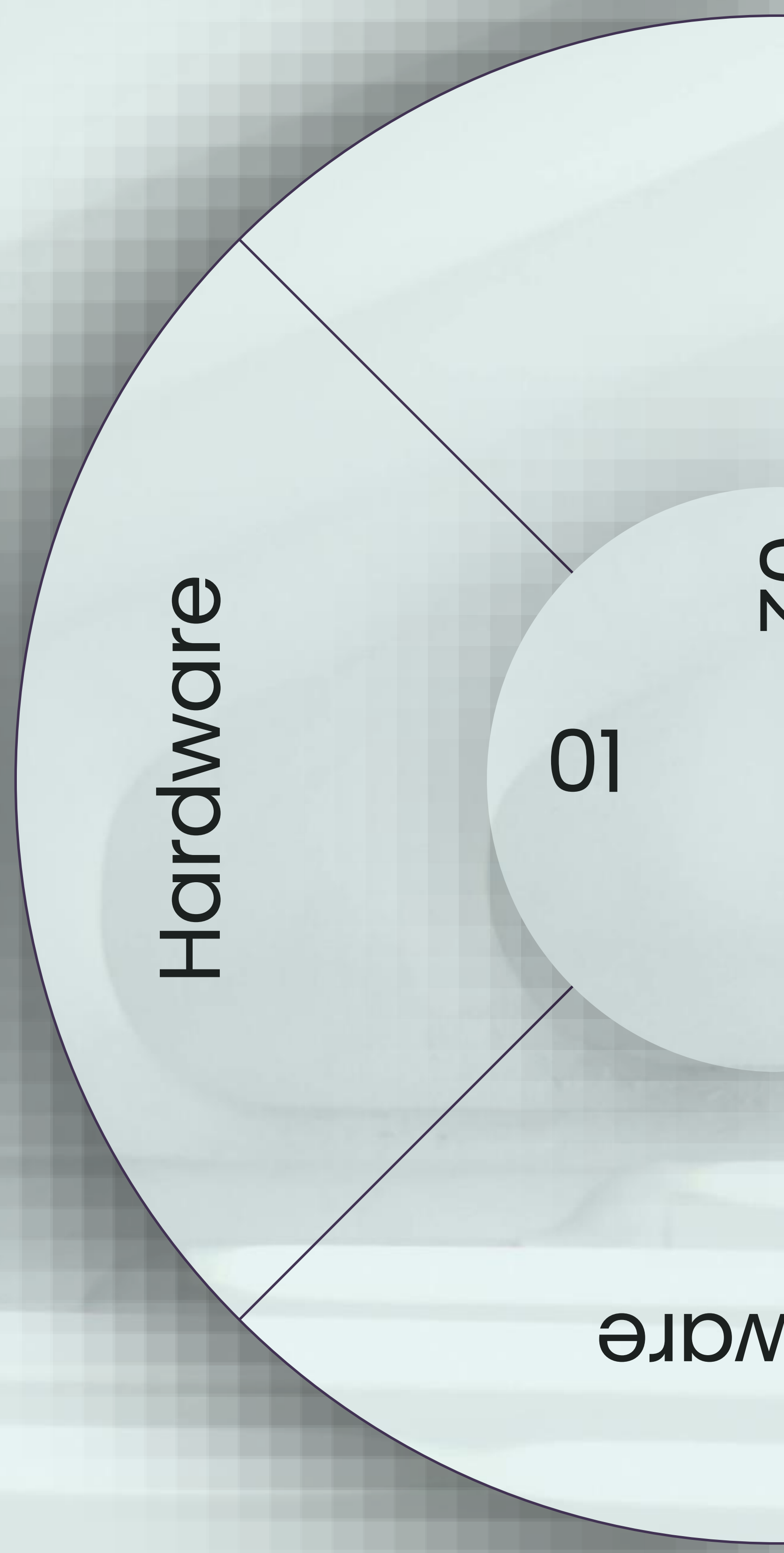
01
Hardware
Microcontroller
Sensor

02
Software
Dataset
ML Models

03
Deployment
Web App

03

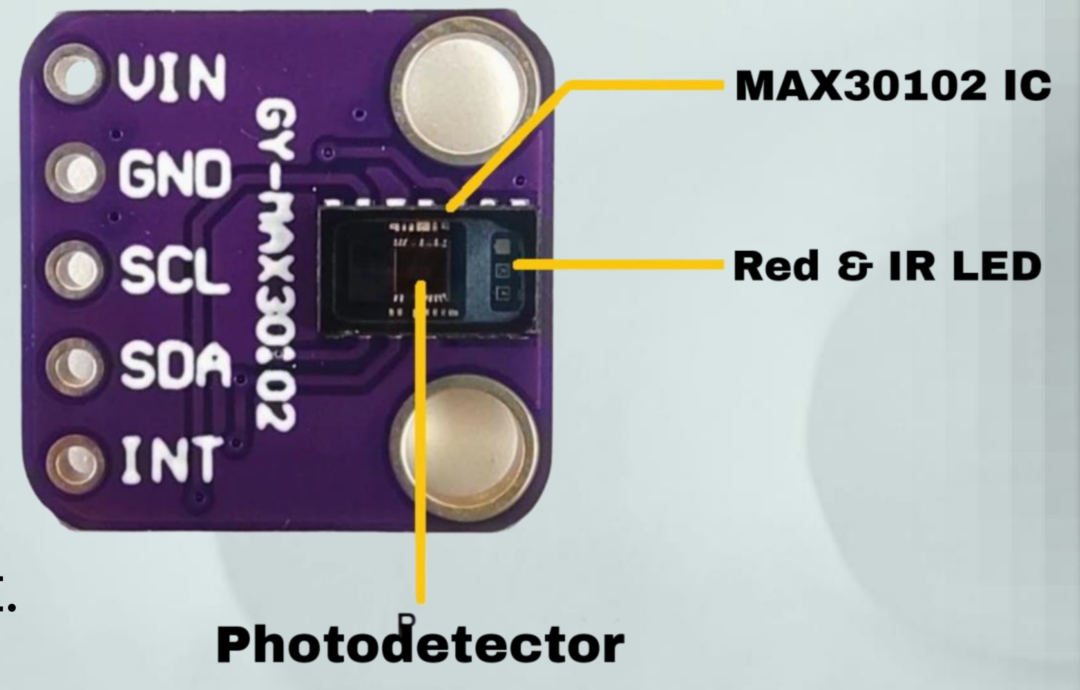
Hardware



Hardware

MAX30102 Sensor

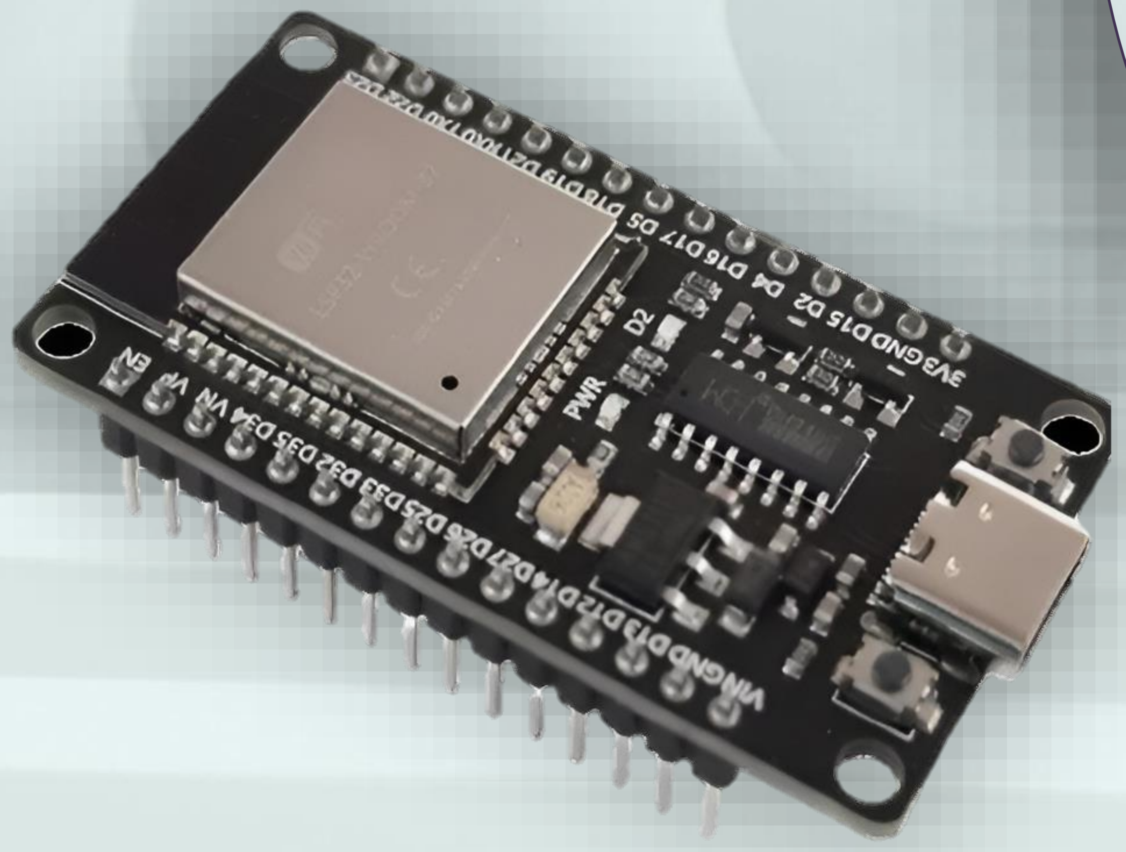
- The **GY-MAX30102 sensor** is a compact optical biosensor designed for capturing the **PPG signals.**
- The sensor includes:
  - **Red LED** with wavelength of **660nm**.
  - **IR LED** with wavelength of **880nm**.
  - **Photodetector** to measure the reflected light.
  - Analog to digital convertor **(ADC)**
  - Digital processing to output the **digital representation of the PPG signal.**

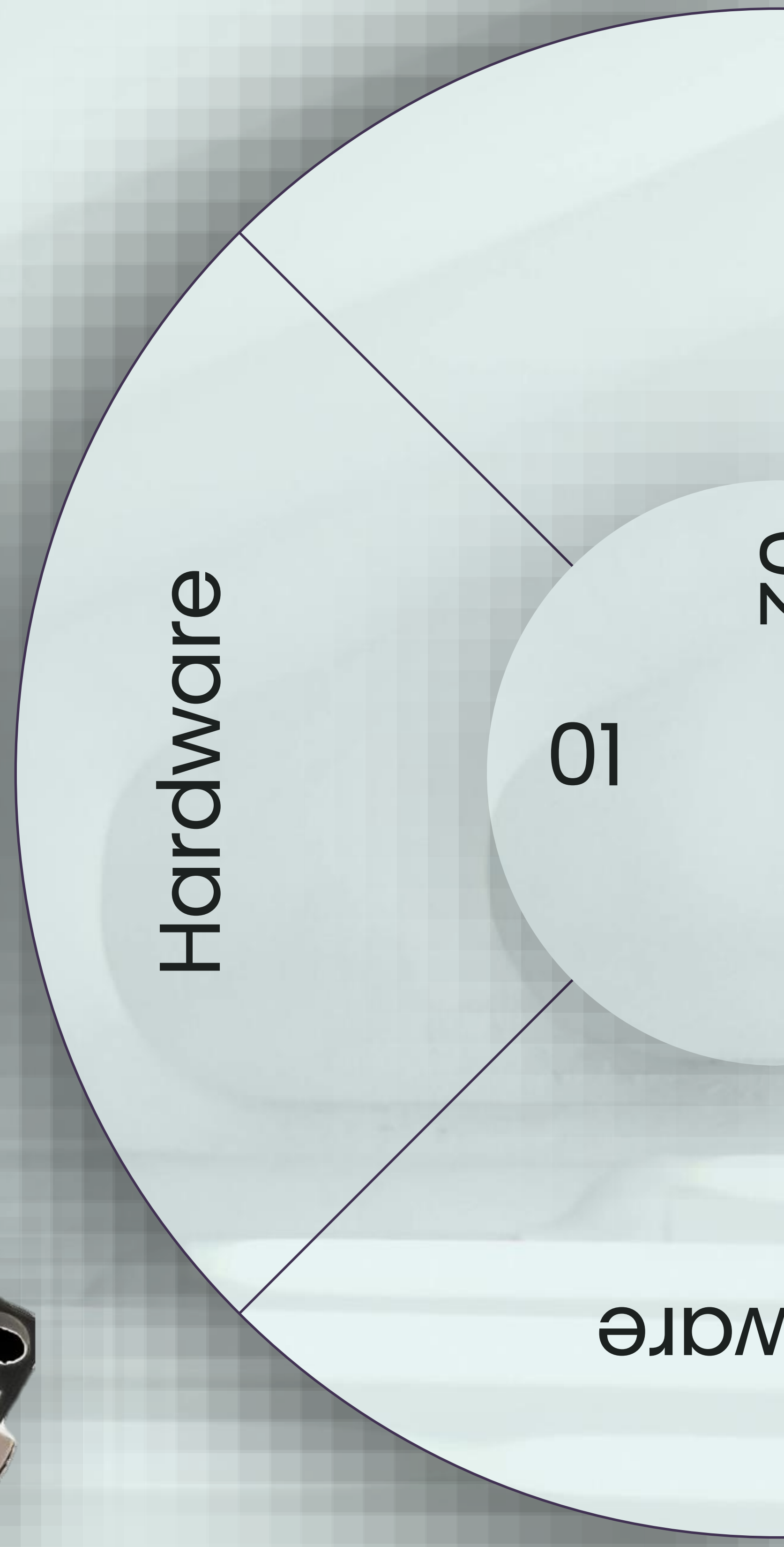


Hardware

ESP23 Microcontroller

- ESP32 to **collect and process** acquired **signals** and sending the data for further analysis.
- Why ESP32 instead of **Arduino** in our project?
  - Built-in **Wi-Fi** and **Bluetooth**.
  - Higher Processing Power.





Hardware

ESP23 + MAX30102

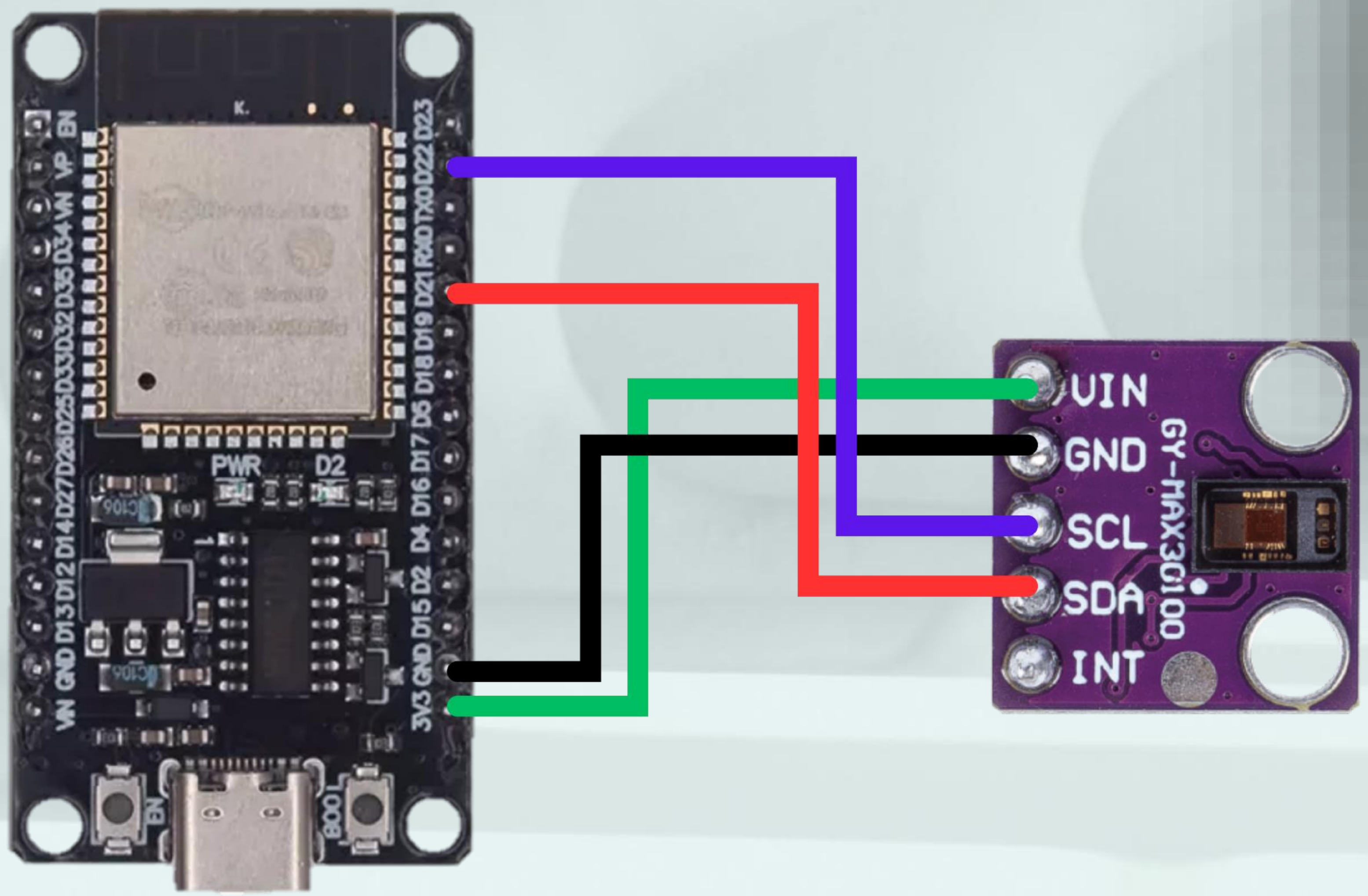
MAX30102 -> ESP32

SCL -> GPIO22
(Clock signal that synchronizes data transfer)

SDA -> GPIO21
(Carries the actual data)

VIN -> 3V3
(3.3v)

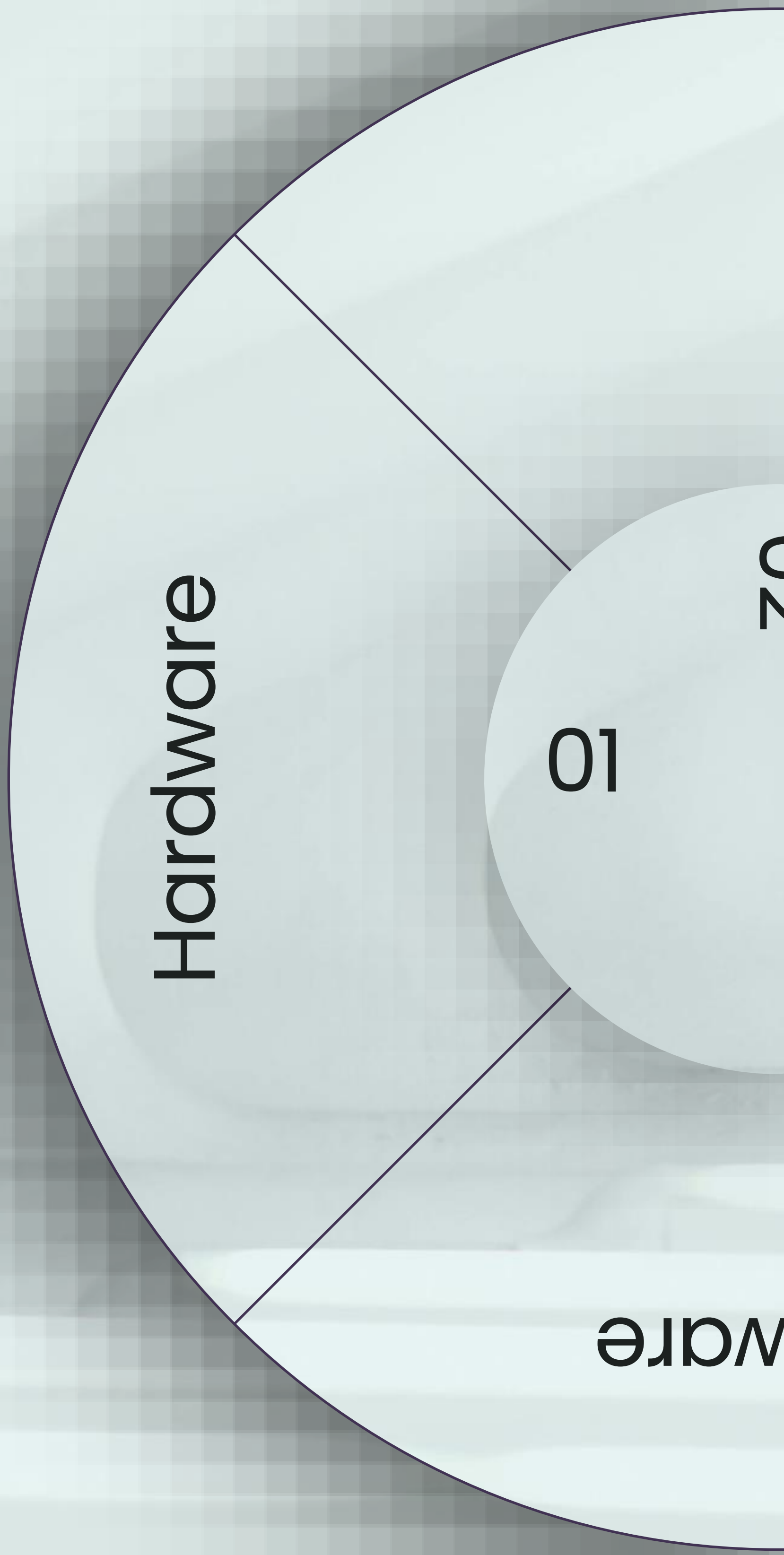
GND -> GND



Hardware

Sensor Limitations

- There are **2** major problems with the signal taken from the signal
  - **External light** making much noise to the signal.
  - **Finger pressure and movement** makes the baseline drift.



Software

Dataset

Software

Public Datasets

- The **absence of public PPG-Glucose datasets is the core problem** of this project.
- Any **public** PPG related datasets **focuses** on tracking basic **vital signs**, such as heart rate or SpO2.
- The **only public dataset** are the Mendeley dataset, published by **researchers** from the University of Science and Technology in Mazandaran, **Iran. But** they used another wavelength (**Green Light**).

Software

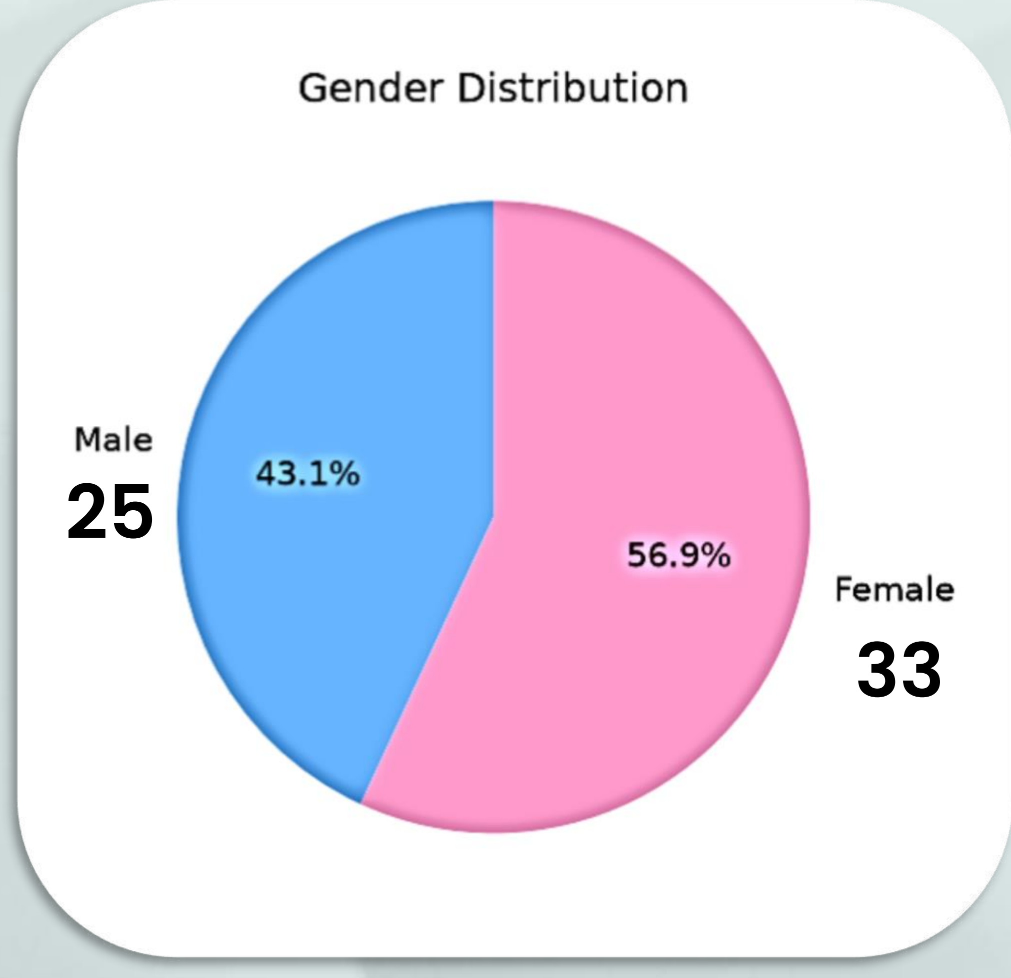
Dataset Acquisition

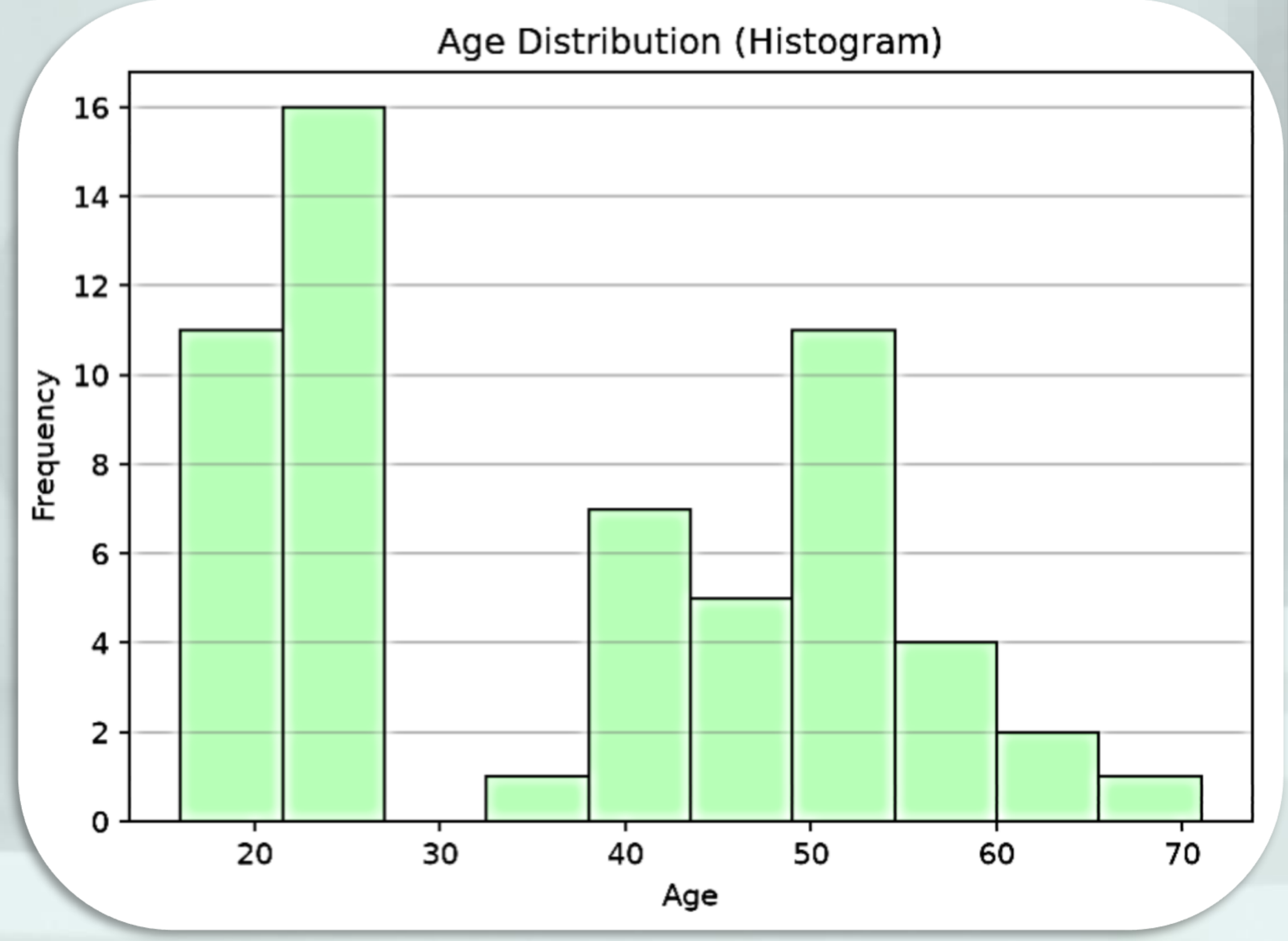
- We were able to collect a total dataset of **58** subjects.
- With the attributes:
  - PPG Signal (**3 -4** mins)
    - **IR** LED
    - **Red** LED
  - Invasive Blood Glucose Level (right after signal taken).
  - Height
  - Weight
  - Age
  - Gender

	A	B	C	D	E	F	G	H
1	time	red	ir	glucose	weight	hight	age	gender
2	80.498	113654	120535	92	113	155	56	f
3	80.576	113743	120650					
4	80.652	113786	120801					
5	80.73	113851	120875					
6	80.806	113871	120951					

Software

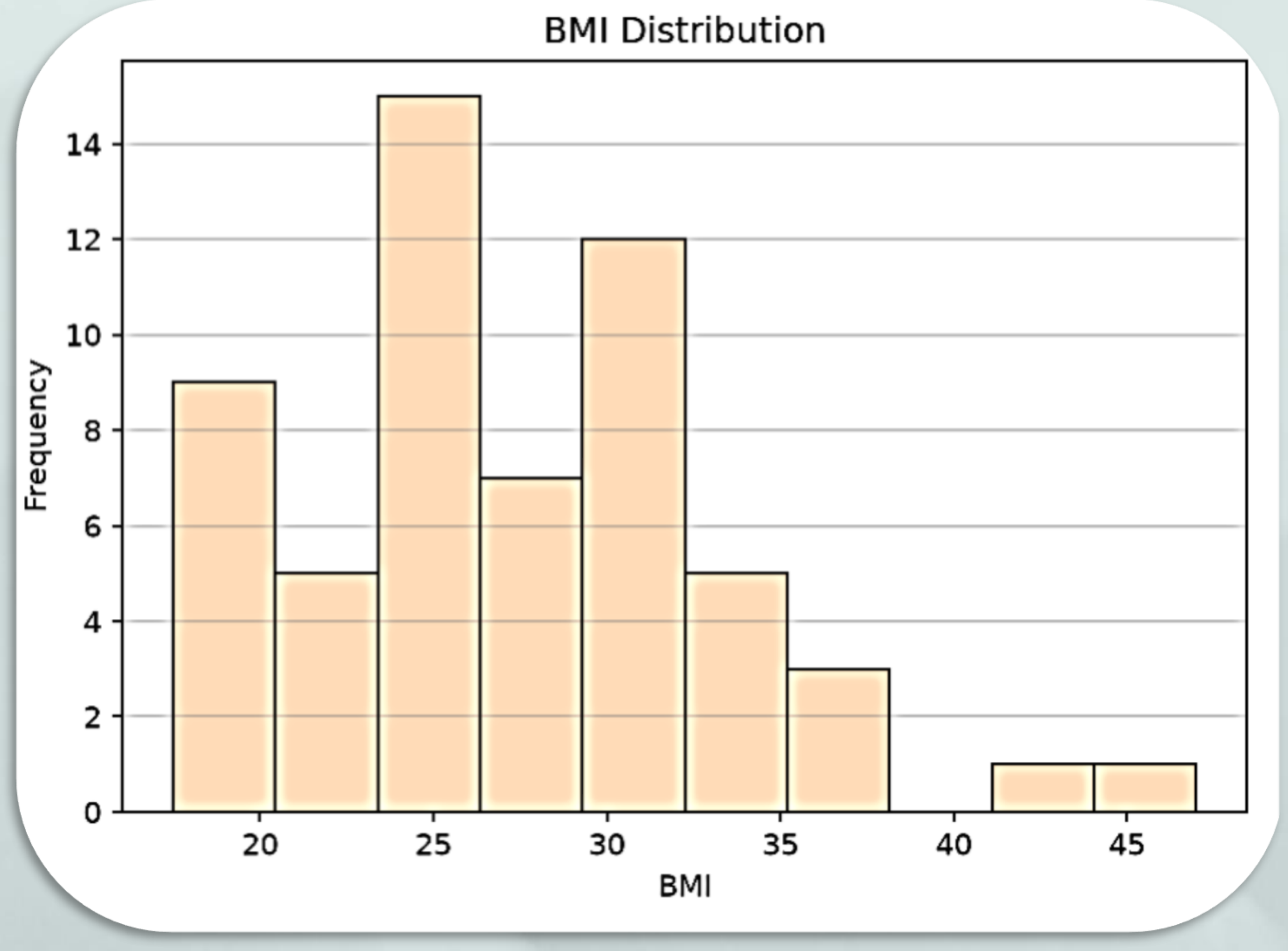
Dataset Acquisition

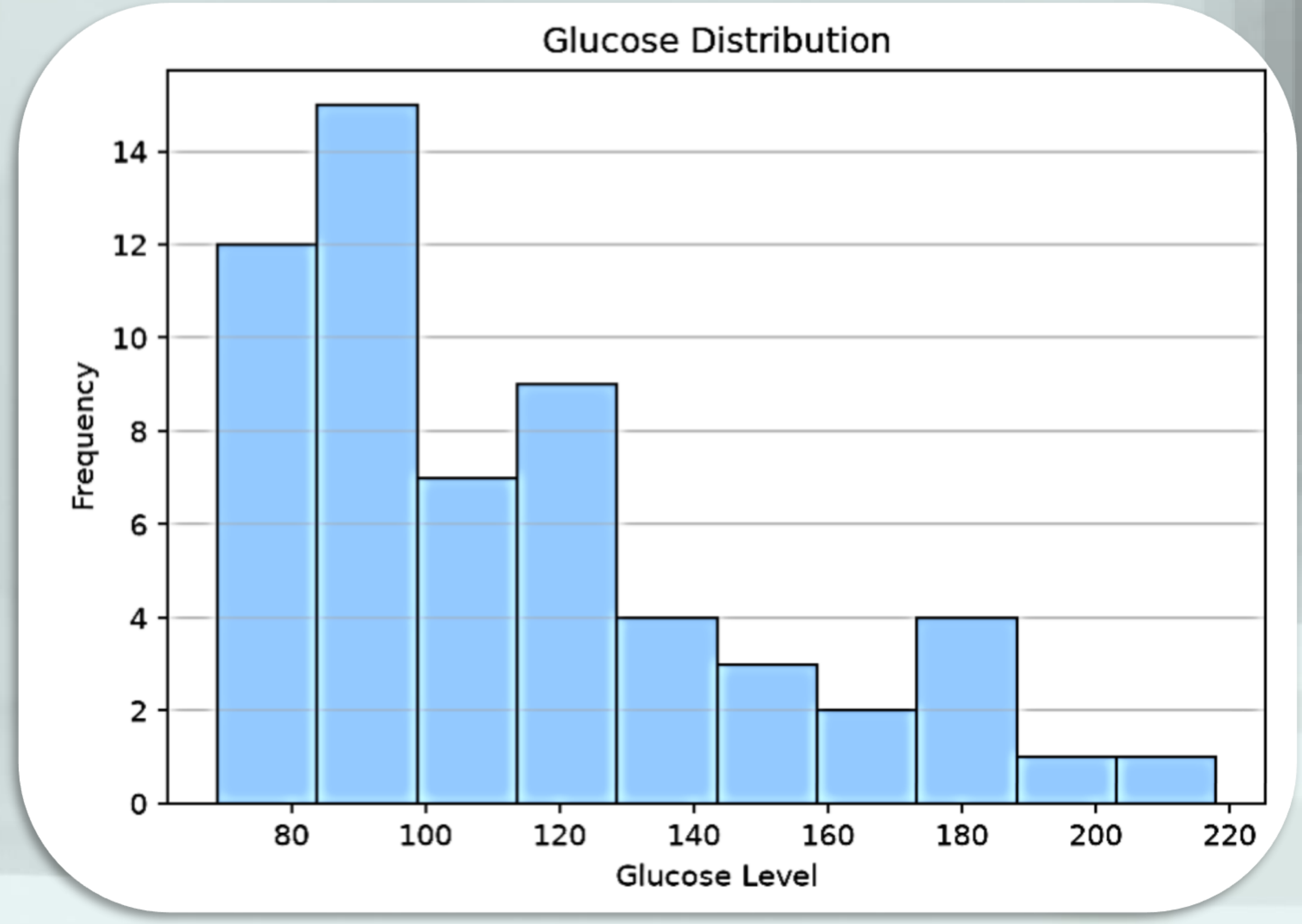




Software

Dataset Acquisition





Preprocessing

Software

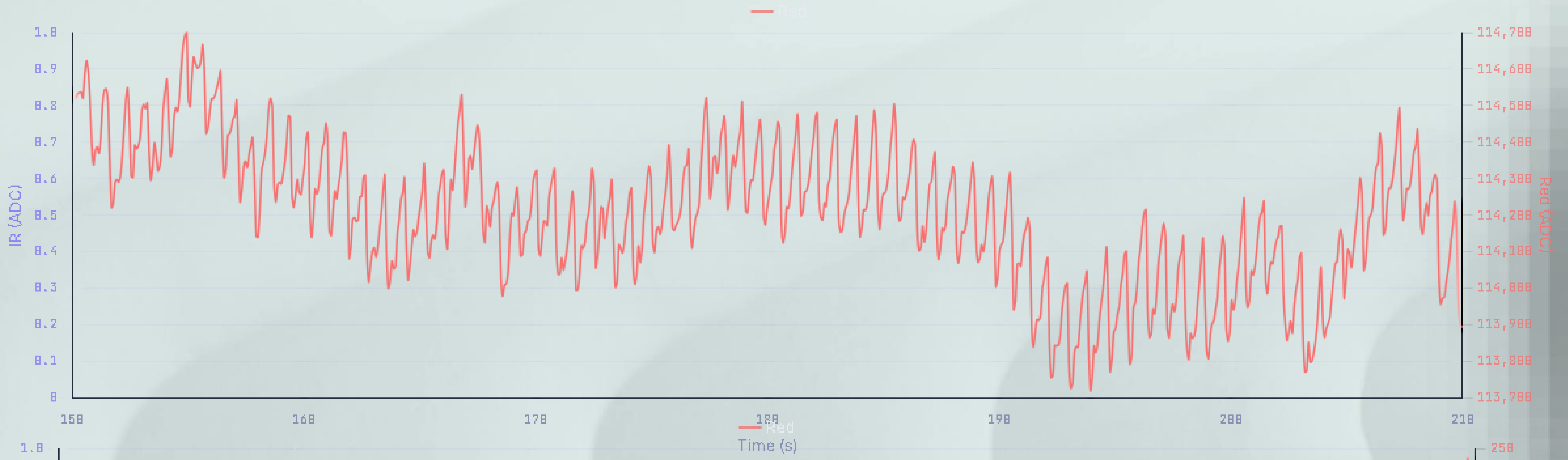
Filtration

- Third-order **Butterworth bandpass** filter with:
  - Lower cutoff frequency = **0.5** Hz
  - Upper cutoff frequency = **8** Hz
- The useful PPG signal information is concentrated in **0.8–3 Hz**
- To remove **electronic noise** (10 -20 Hz) and any **noise** caused by:
  - **Breathing effects.**
  - **Body posture changes.**
  - **Baseline drift** (finger pressure).

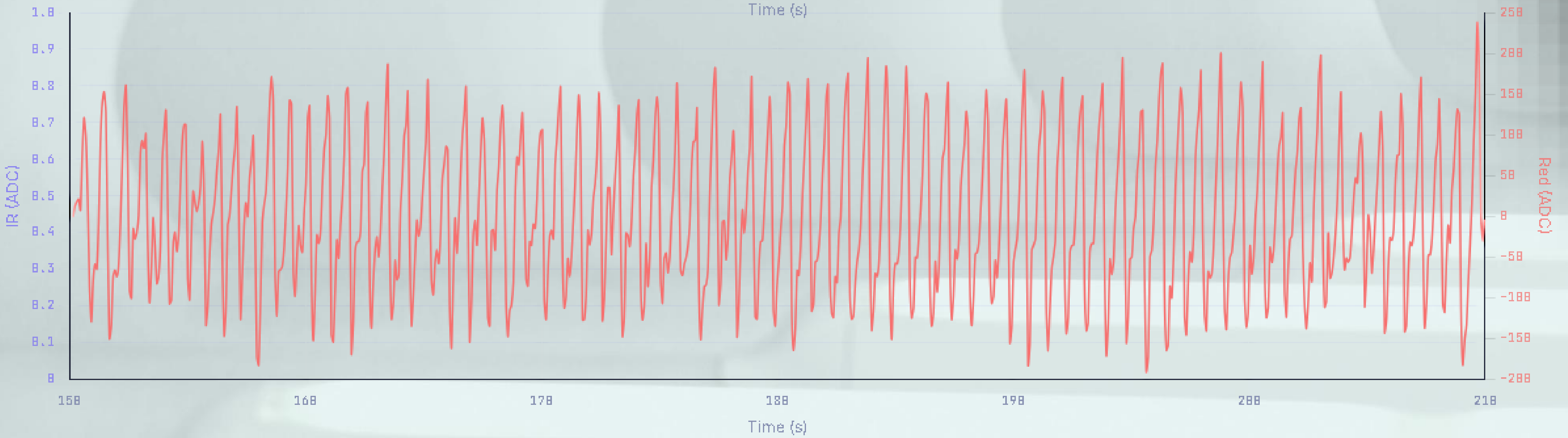
Software

Filtration

- Before



- After



Segmentation

Software

Segmentation

- A **5** different segmentation pipelines approaches were used:
  - **3** for **machine learning**.
  - **2** for **deep learning**.

Software

Segmentation

- **3 Machine Learning**.

01 Full Wave

- Initial baseline method.
- The **whole signal** is used for feature extraction.

02 Single Minute

- Finding the single "**most perfect**" continuous 1 min window to use for feature extraction.

03 1 Min Multi-Window

- Slicing the signal into 1 min windows.
- Take the **mean** after feature extraction.

Software

Segmentation

- **2 Deep Learning**

04

10s Segments

- 10s windows with **50% overlap**.
- The median and standard deviation are computed globally.
- Segments that **more than 4σ** from the median is **discarded**.

05

1s Heartbeat Segments

- Using the **NeuroKit2** library to find heartbeat peaks.
- Every peak found we extract **50** point before and after it.
- Same outlier method as (04).

Feature Extraction

Software

Features

1. **BMI** (Body Mass Index):
    - $BMI = \frac{weight\ (kg)}{[height\ (m)]^2}$
2. **DC** Component (Direct Current):
    - The **mean** value of the raw PPG signal.
    - Represents everything that remains relatively constant (Skin, Bone, Non-pulsatile tissues).
3. **AC** Component (Alternating Current):
    - The pulsatile part of the signal caused by heartbeats.
    - The (**max – min**) of the filtered signal.

Software

Features

4. **Ratio R** (Ratio of Ratios):
    - Ratio of the **AC** to **DC** amplitude of the **Red** wavelength and dividing it by that of the **IR** wavelength.
    - Used for **SpO_2** estimation ($SpO2=110-25R$).
5. **Slope**
    - The average **speed of rise** of the waveform.

Software

Features

6. **Skewness**
    - The **asymmetry** of the waveform.
7. **Kurtosis**
    - How **sharp** or **flat** the peaks are.
8. **Signal Power**
    - Total signal **energy**.
    - The mean squared amplitude of the signal.

Models

Software

ML Models

Model	MAE (mg/dL)	RSME (mg/dL)	R^2	Segmentation
XGBoost	9.98	12.91	0.87	01
CatBoost	13.18	16.37	0.79	01
KNN	16.03	18.36	0.74	03
Random Forest	17.21	22.35	0.62	02
Linear Regression	20.05	24.84	0.52	01
SVR	17.67	28.19	0.39	02

Research Paper	Publish Year	Model	MAE	RMSE	R^2
Photoplethysmography based non-invasive blood glucose estimation using systolic-diastolic framing MFCC features and machine learning regression	2025	SVR	4.014	26.01	Not reported
Non-Invasive PPG-Based Estimation of Blood Glucose Level	2023	Support Vector Machine (SVM) / Random Forest	11.86	22.5	Not reported
Blood Glucose Level Estimation Using Photoplethysmography (PPG) Signals with Explainable Artificial Intelligence Techniques	2024	CatBoost Regressor	25.16	39.13	0.71

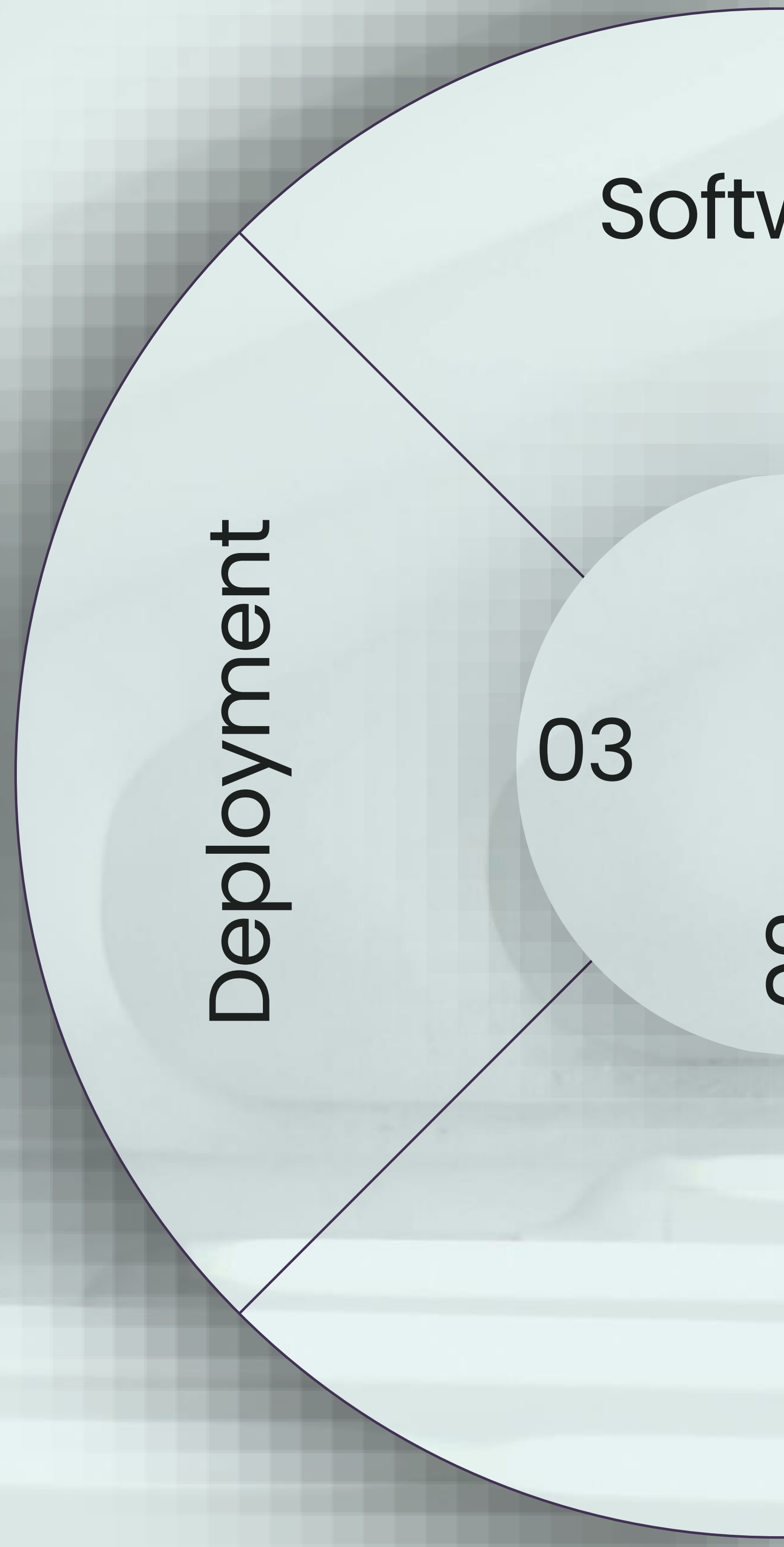
Software

DL Models

Model	MAE (mg/dL)	RSME (mg/dL)	R^2	Segmentation
Two-headed Resnet	22.57	28.42	0.34	05
Resnet-1d	21.96	29.24	0.30	05
Hybrid CNN-BiLSTM	23.41	30.19	0.26	05

Research Paper	Publish Year	Model	MAE	RMSE	R^2
Non-invasive blood glucose monitoring using PPG signals with various deep learning models and implementation using TinyML	2025	ResNet34 with customized 1D Convolution (Conv1D) layers	18.54	27.48	0.40

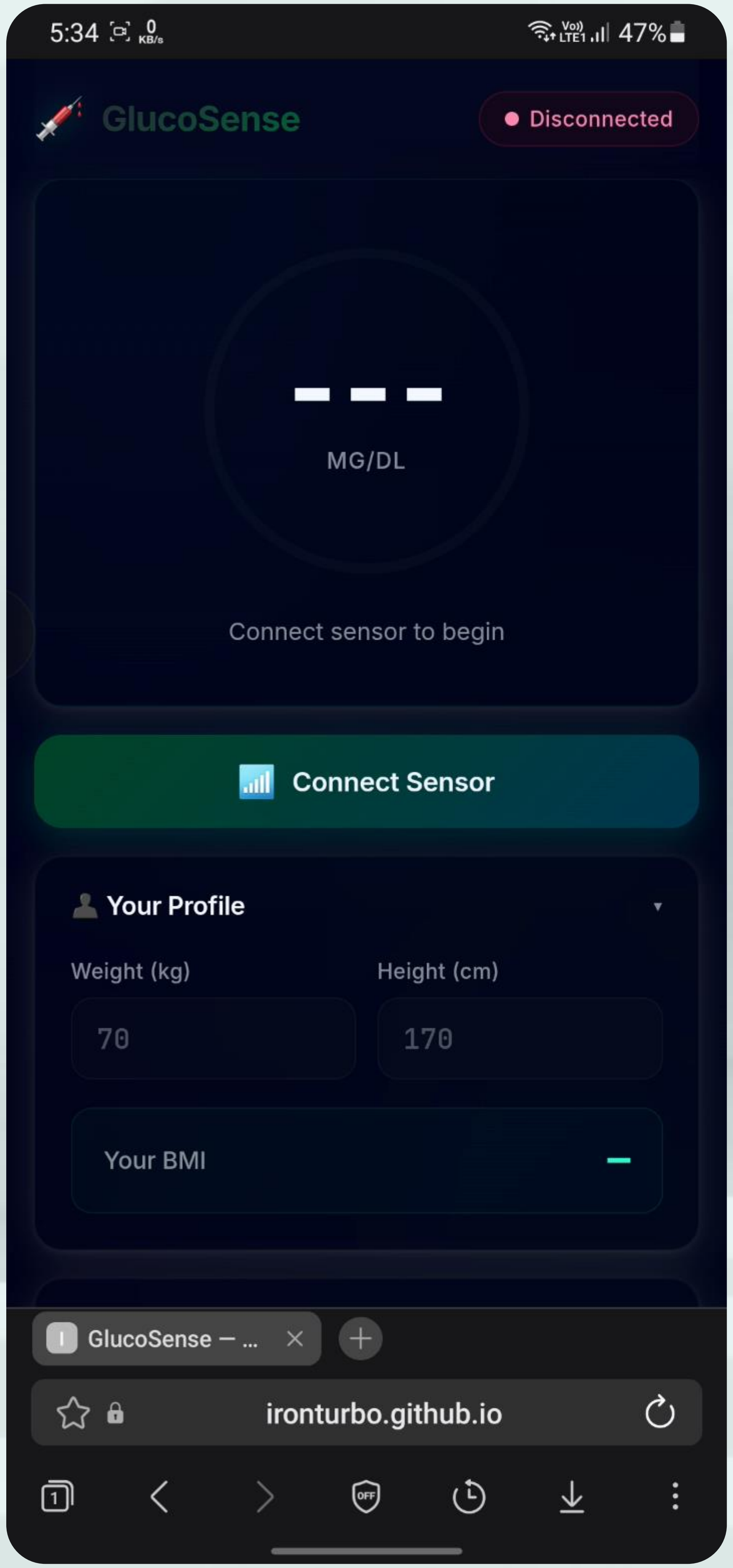
Deployment

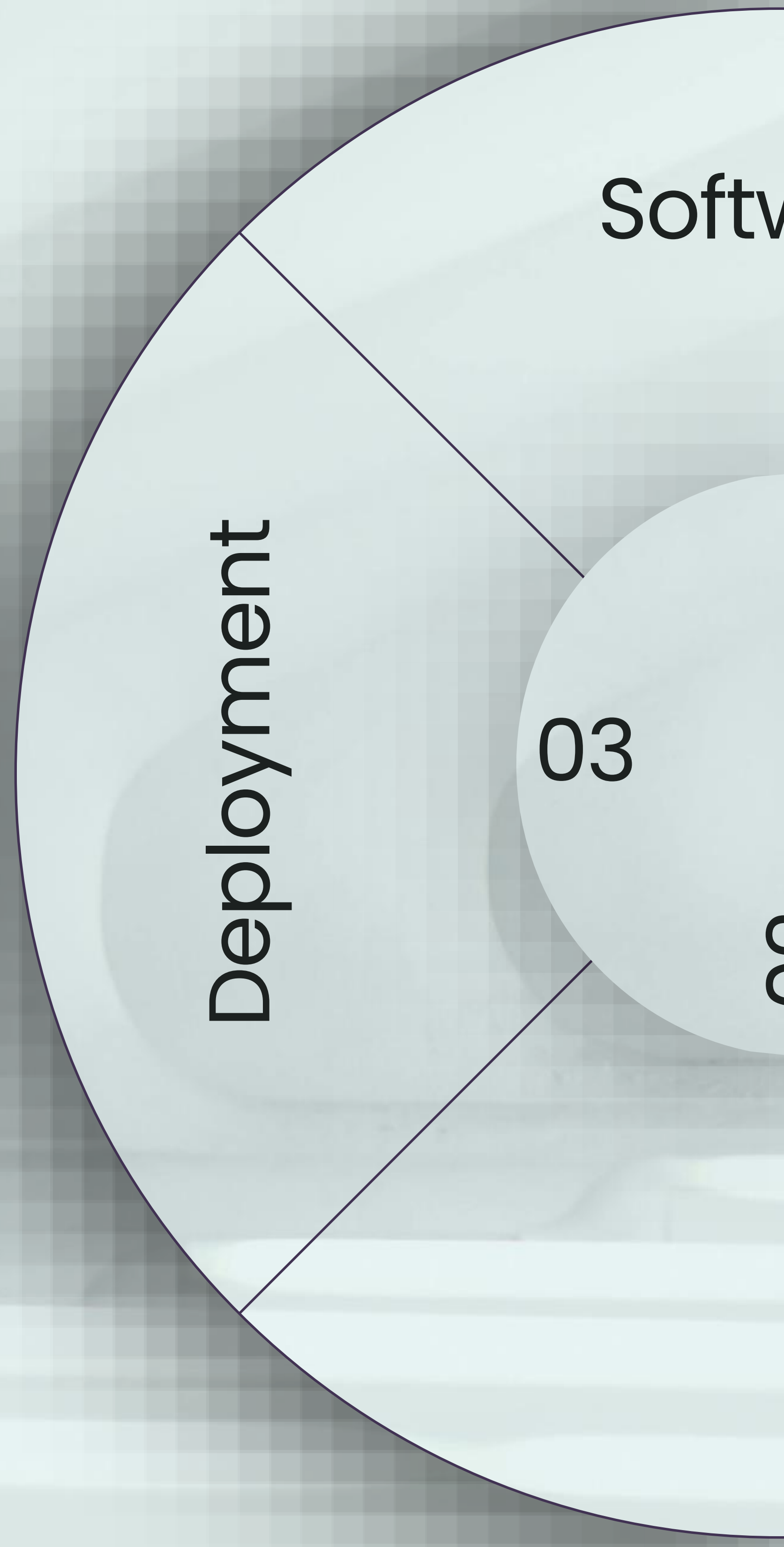


Deployment

Web App

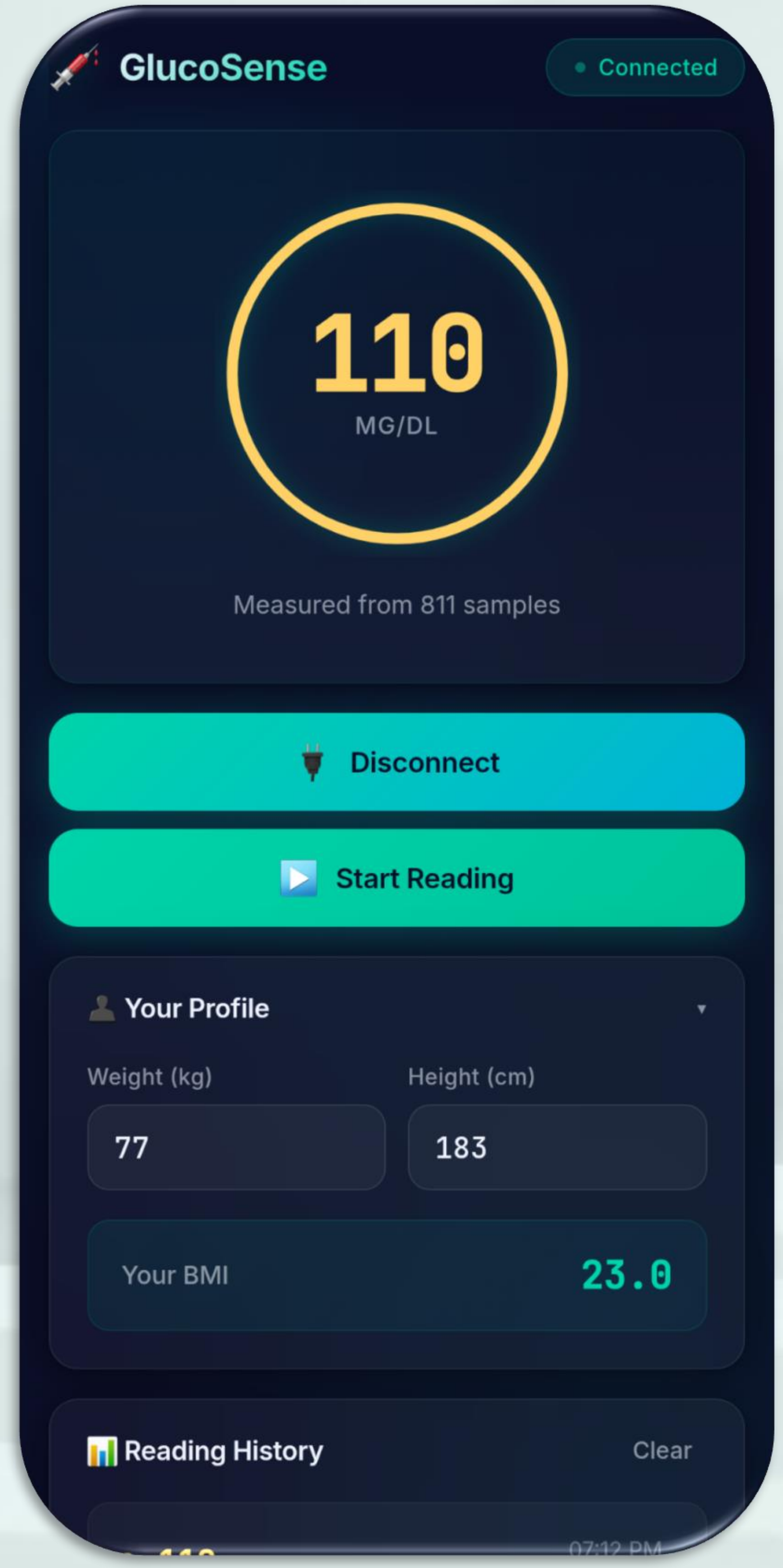
- A web application (Progressive Web App - **PWA**) runs directly in the user's **browser** and communicates with the ESP32 through the Web **Bluetooth** API without the need of a **backend server**.
- The machine learning model is also embedded inside the web application as **JavaScript** code.

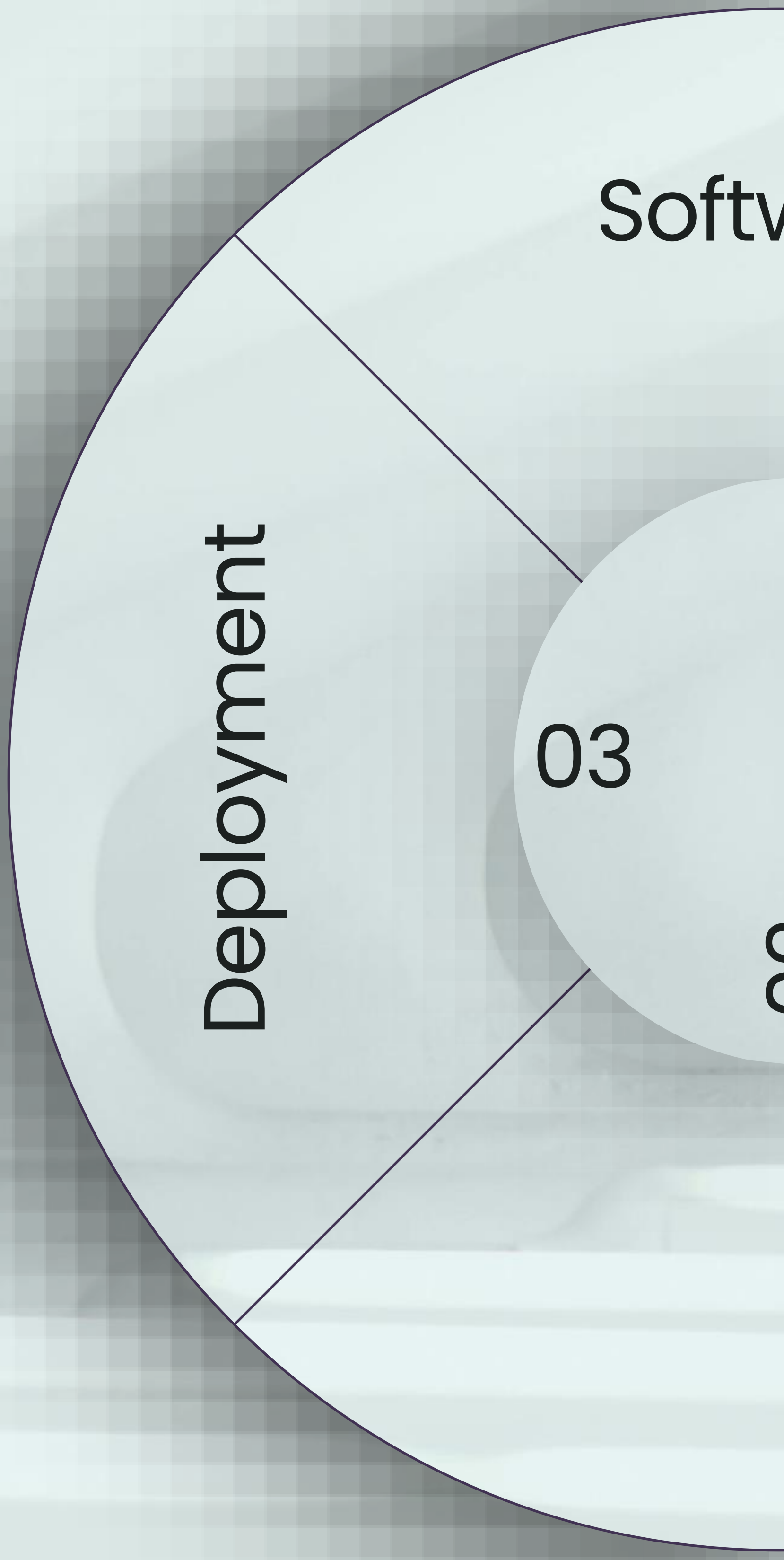




Deployment

Web App





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Thank You!

Any Questions?